not emit energy in a smooth, continuous wave, but in measured amounts, or packets, later called "quanta" (from the Latin word *quantum*, meaning "how much"). A quantum is the smallest possible packet of energy.

Planck's radiation law

The theory, later called Planck's radiation law, finally explained the relationship between the temperature of an object and the energy emitted from that object in the form of electromagnetic radiation. Planck expressed this relationship as a mathematical equation: $E = h\nu$. The energy (E) in a photon equals its electromagnetic radiation frequency (ν) multiplied by Planck's constant (h).

In 1900, Planck presented his theory of light as "quantized" energy packets to the German Physical Society. Overturning all past physical theory, it helped initiate a revolution in physics, and today it is regarded as the origin of quantum theory. In 1905, Planck's hypothesis was verified by Albert Einstein, when he extended it in order to explain the photoelectric effect: the existence of discrete energy packets during the transmission of light.

Defining his constant h allowed Planck to devise a new set of physical units. Among these are the Planck length, the smallest unit of measurement possible: 1.6×10^{-35} meters. The amount of time it takes for a photon to travel a Planck length at the speed of light is one unit of Planck time. This is the smallest measurable unit of time: 5×10^{-43} seconds.

***Planck's theory**, based on the bold notion that energy was available only in fixed "packets" (quanta), paved the way for modern technologies such as solar power.*

HIS **QUANTUM THEORY** IS **1** OF THE **2 FOUNDATION STONES** OF 20TH-CENTURY SCIENCE

NOMINATED FOR THE **NOBEL PRIZE** IN **PHYSICS 74** TIMES

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